1RV23AI092

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Academic Year: 2025-2026 Even

Department of Artificial Intelligence and Machine Learning

Date: 15/04/2026	Test - 1	Max. Marks : 10 + 50
Semester: VI	UG	Duration : 120 mins
Course Title: Generative Artificial Intelligence		Course Code: AI365TDD

Part - A

Q. No	Question	M	BT	CO
1.1	In a Variational Autoencoder (VAE), why do we learn a distribution (mean and variance) instead of a single latent vector?	2	2	1
1.2	Define the probability functions used in discriminative and generative models.	2	1	1
1.3	List any two limitations of Large Language models (LLMs).	2	1	3
1.4	What is the role of the embedding layer in Large Language Models?	2	2	1
1.5	In Variational Autoencoders (VAEs), why is the loss function defined as a combination of two terms instead of a single objective?	2	2	1

Part - B

Q. No	Question	M	BT	CO
1	Construct a Transformer-based encoder–decoder model that uses: • Self-attention • Masking • Positional encoding Draw a neat architecture diagram and explain each component involved in encoding and decoding the input sequence.	10	4	2
2	An autoencoder is used to compress image data through its encoder. Consider an input image of size **28 × 28 × 1**. The encoder consists of convolutional layers with the following parameters: Layer 1: 16 filters, kernel = 3×3, stride = 2, padding = 1 Layer 2: 32 filters, kernel = 3×3, stride = 2, padding = 1 Write a Python program (using PyTorch or TensorFlow) to implement the encoder.	10	3	2

3a	How does representation learning simplify operations on data (such as modifying attributes) compared to directly manipulating input space? Explain with a suitable example.	5	4	3
3b	Implement latent space sampling for a Variational Autoencoder using the reparameterization trick, without directly sampling from a normal distribution at the output layer. Your answer should include: • Intermediate transformations steps • Architecture block diagram • Graphical illustration of sampling	5	3	2
4a	Autoencoders are often used as denoising models by learning compact latent representations. (i) Explain how an autoencoder removes noise during training. (ii) Analyze the effect of the following on reconstruction quality: • A very small latent space • A very large latent space	5	3	3
4b	An encoder–decoder model is trained to reconstruct input data. Another variant of the model introduces a constraint on the latent space distribution. (i) How would you evaluate the performance of both models during training? (ii) Explain the components of their loss functions used in each case. (iii) Write a Python snippet to compute the reconstruction loss for the first model.	5	3	2
5	An encoder–decoder model maps each input to a fixed latent vector and achieves high reconstruction accuracy but poor generalization. (i). Identify this class of models. (ii). Explain how introducing a distribution over latent variables modifies the objective function. Provide the mathematical formulation. (iii). List two differences between this model and its probabilistic variant (Variational Autoencoder).	10	4	4